

1 Solution — GIAM 2.7

Problem 5

1.1 Problem

Read the following proof that the sum of two odd numbers is even. Discuss the rules of inference used.

Proof: Let x and y be odd numbers. Then $x = 2k + 1$ and $y = 2j + 1$ for some integers j and k . By algebra, $x + y = 2k + 1 + 2j + 1 = 2(k + j + 1)$. Note that $k + j + 1$ is an integer because k and j are integers. Hence $x + y$ is even. **Q.E.D.**

1.2 The Rules, Line by Line

A four-sentence proof, with every inference named. The sentences are English; the right column is the machinery.

the proof, line by line	rule of inference
"Let x and y be odd numbers."	set up universal generalisation <i>x, y arbitrary; also the hypothesis of a conditional proof</i>
"Then $x = 2k+1$ and $y = 2j+1$ for some integers j and k ."	universal modus ponens, twice <i>definition of "odd" applied to x and to y, then existential instantiation names k and j</i>
"By algebra, $x + y = 2k+1+2j+1 = 2(k+j+1)$."	substitution and transitivity of = <i>not a logical rule — the arithmetic axioms</i>
"Note that $k+j+1$ is an integer because k and j are integers."	universal modus ponens <i>closure of $\mathbb{Z}$ under addition</i>
"Hence $x + y$ is even."	existential generalisation, then the definition of "even" <i>$\exists m \in \mathbb{Z}$ with $x+y = 2m$, so $x+y$ is even</i>
"Q.E.D."	universal generalisation — discharge <i>x and y were arbitrary, so this holds for all odd x, y</i>

The two definitions are used in opposite directions		
unpacking odd :	$n \text{ is odd} \Rightarrow \exists m \in \mathbb{Z}, n = 2m+1$	<i>forward, at the start</i>
packing even :	$\exists m \in \mathbb{Z}, n = 2m \Rightarrow n \text{ is even}$	<i>backward, at the end</i>
<i>Legitimate because a definition is a biconditional — it may be run either way.</i>		

The overall shape		
assume the hypothesis for arbitrary $x, y \rightarrow$ unpack definitions $\rightarrow$ compute $\rightarrow$ repack $\rightarrow$ generalise		
<i>Almost every direct proof in mathematics has exactly this skeleton.</i>		

Figure 1.1: A two-column annotation of the proof. Line one, ‘Let x and y be odd numbers’: sets up universal generalisation — x and y are arbitrary, and this is also the hypothesis of a conditional proof. Line two, ‘Then $x = 2k+1$ and $y = 2j+1$ for some integers j and k ’: universal modus ponens applied twice, using the definition of odd on x and on y , followed by existential instantiation to name k and j . Line three, ‘By algebra, $x+y = 2k+1+2j+1 = 2(k+j+1)$ ’: substitution and transitivity of equality — not a logical rule but the arithmetic axioms. Line four, ‘Note that $k+j+1$ is an integer because k and j are integers’: universal modus ponens, using closure of the integers under addition. Line five, ‘Hence $x+y$ is even’: existential generalisation followed by the definition of even, since there exists an integer m with $x+y$ equal to $2m$. Line six, ‘Q.E.D.’: universal generalisation, discharging the assumption, since x and y were arbitrary. A panel notes the two definitions are used in opposite directions — odd is unpacked forward at the start, even is packed backward at the end — which is legitimate because a definition is a biconditional. A final panel gives the overall shape: assume the hypothesis for arbitrary x and y , unpack definitions, compute, repack, generalise.

line of the proof	rule of inference
“Let x and y be odd numbers.”	Sets up universal generalisation : x, y are <i>arbitrary</i> . Simultaneously the hypothesis of a conditional proof .
“Then $x = 2k + 1$ and $y = 2j + 1 \dots$ ”	Universal modus ponens , twice — the definition $\forall n, n \text{ odd} \Rightarrow \exists m \in \mathbb{Z}, n = 2m + 1$ applied at x and at y — followed by existential instantiation to give the witnesses the names k and j .
“By algebra, $x + y = 2k + 1 + 2j + 1 = 2(k + j + 1)$.”	Substitution of equals for equals, plus transitivity of $=$. Not a rule of logic — these are the arithmetic axioms doing the work.
“ $k + j + 1$ is an integer because k and j are.”	Universal modus ponens on the closure property $\forall a, b \in \mathbb{Z}, a + b \in \mathbb{Z}$.
“Hence $x + y$ is even.”	Existential generalisation — from the specific witness $k + j + 1$ to $\exists m \in \mathbb{Z}, x + y = 2m$ — then the definition of <i>even</i> run backwards.

“Q.E.D.” **Universal generalisation:** since x and y were arbitrary, the conclusion holds for *all* odd x, y . This discharges the opening assumption.

Table 1.1

The algebra was verified on 200 random integer pairs, and the theorem itself on all 2704 pairs of odd numbers in $[-51, 51]$.

1.3 Remark — The Quantifier Rules Do the Real Work

The propositional rules of Section 2.6 barely appear here. What actually drives the proof are the four **quantifier rules**:

rule	form	what it does here
universal instantiation	$\forall x P(x) \vdash P(c)$	apply a definition to a particular thing
existential instantiation	$\exists x P(x) \vdash P(k)$, new name k	name the witness
existential generalisation	$P(c) \vdash \exists x P(x)$	forget which witness it was
universal generalisation	$P(x)$ for arbitrary $x \vdash \forall x P(x)$	discharge at the end

Table 1.2

The proof instantiates twice at the start, generalises twice at the end, and the middle is arithmetic. That in–out sandwich is the standard architecture of a direct proof, and recognising it makes unfamiliar proofs far easier to read.

1.4 Remark — “Let x Be Arbitrary” Is a Promise You Must Keep

The opening line does double duty, and both jobs matter. It **assumes the hypothesis** (that x, y are odd), which is what makes the final statement a conditional. And it declares x, y **arbitrary**, which is what licenses the closing generalisation.

The second is the fragile one. Universal generalisation is valid only if *nothing beyond the stated hypothesis* was assumed about x and y . Had the proof said “let $x = 3$ ” or slipped in “and suppose $x < y$,” the final “for all” would be unearned. Notice the proof is scrupulous here: after naming k and j it never uses any property of them except that they are integers. This is the same “let x be arbitrary ... therefore for all x ” pattern identified in Problem 3, now seen doing real work.

1.5 Remark — Definitions Are Biconditionals, So They Run Both Ways

The proof uses the definition of *odd* in the forward direction and the definition of *even* in the backward direction:

$$\underbrace{n \text{ odd} \Rightarrow \exists m, n = 2m + 1}_{\text{unpacking, at the start}}, \quad \underbrace{\exists m, n = 2m \Rightarrow n \text{ even}}_{\text{packing, at the end}}.$$

This asymmetry is deliberate and is the shape of most proofs: **unpack the hypothesis into raw arithmetic, do the computation, then repack the result into the vocabulary of the conclusion.** It is legitimate because a definition is an *if and only if* — unlike an ordinary theorem, whose converse must be proved separately (Problem 2.2.9). Writing “ x is odd, so $x = 2k + 1$ ” and “ $x + y = 2m$, so $x + y$ is even” both invoke the same biconditional, read in opposite directions.

1.6 Remark — Distinct Names for Distinct Witnesses

A subtle but essential point: the proof writes $x = 2k + 1$ and $y = 2j + 1$ with **two different letters**. Existential instantiation must introduce a *fresh* name each time. Had the proof written $x = 2k + 1$ and $y = 2k + 1$, it would have silently assumed $x = y$ and proved only that a number’s double is even — a far weaker claim.

This is the same distinctness care that Problem 2.5.1 required when negating $\exists!$: bound variables are not automatically different, so if you need them distinct you must arrange it. Here the discipline is to pick an unused letter every time you name a witness.

1.7 Remark — Not Every Step Is a Rule of Logic

The line “by algebra” is worth flagging, because it is *not* an inference rule at all. It appeals to the axioms of arithmetic — associativity, commutativity, distributivity — plus substitution of equals for equals. Likewise “ $k + j + 1$ is an integer” appeals to a **closure property** of $\mathbb{Z}$, a fact about the number system rather than about logic.

A complete formalisation would list these as additional non-logical premises. Ordinary mathematical writing suppresses them because they are universally agreed, which is exactly why an exercise like this is useful: it reveals how much a two-word phrase like “by algebra” is actually carrying. The step deserves scrutiny precisely because it is where a plausible-looking proof can hide an error.

1.8 Remark — Why the Closure Step Is Not Pedantry

Newcomers often find “note that $k + j + 1$ is an integer” fussy — of course it is. But the step is genuinely required, because the definition of *even* demands $n = 2m$ **for an integer m** . Producing the equation $x + y = 2(k + j + 1)$ is not enough on its own; one must confirm that the thing in parentheses lives in the right set.

The necessity becomes obvious if the setting changes. Over the integers $7 = 2 \cdot 3.5$ does *not* make 7 even, because $3.5 \notin \mathbb{Z}$. Every “ n is divisible by d ” claim carries the same hidden requirement, and forgetting to check it is a standard source of bogus proofs. The proof under discussion is careful, and the line is there for a reason.